

15.2 Project management software

Types of project management software

There are many different types of project management software. Many examples of project management software can be used on different devices.

Examples of project management software include Jira, Wrike, Miro and Monday. Microsoft also produces project management software called Microsoft Project.

Project management software is often run on a desktop computer or laptop. However, some projects may need project management software to run on mobile devices such as smart phones or tablets.

Some project management software is web-based. This allows management of projects across the globe. Web-based project management software means that people from different countries can log in and update projects.

DISCUSSION POINT

What projects would benefit from:

- mobile project management software?
- web-based project management software?

Project management software often has different **licensing** options. Personal licences allow you to use the software on a home computer. You cannot use the software for work purposes for a business or organisation. This could be useful if you have a project at home, such as building a shed in the garden.

Single-user licences can be used within an organisation or business. However, they only allow one person to use the software at once. Collaborative project management software allows many people to log in to the software at the same time. This will allow many people to use the project management software at the same time. This is useful if you have large projects with many project managers and stakeholders.

Uses of project management software for supporting projects

Planning

Project management software can be used to schedule project-based tasks to meet milestones and to allocate resources to each task. Project management software can link documents relating to the project such as budgets, goals, requirements and resources to specific tasks. Users will then have access to the data and information that they require to complete each task on schedule and within budget.

Project management software provides users with templates for the planning stage. Templates can be used to set up initial project plans and budgets. Templates from previous

successful projects may be repurposed for the current project. Using project templates allows all projects in an organisation to have common, collaborative and recognisable structures.

Scheduling of tasks

Tasks are jobs that are completed as part of the project. Project management software can be used to schedule and track the progress of tasks. Within project management software, a Gantt chart can be used to show an overview of the tasks required to complete a project. The Gantt chart shows this on a timeline. Gantt charts allow a project manager to assign numbers of hours or days to each task. Each task can also be given a deadline for completion. Gantt charts often sort or group tasks automatically.

The project manager can use the project management software to identify which tasks are dependent on other tasks. Dependent means that a task cannot start until another task has been completed. Tasks can be delegated to different members of the project team. Tasks are given a priority and can be sorted from most important to least important. A priority list helps identify which task should be completed first.

Project management software also allows:

- a project manager or team member to record how many hours are spent on each task
- each task to be marked as complete
- milestones to be created for crucial points of the project
- highlighting of key project parts
- a calendar to be produced of events, milestones and progress achieved.

Allocation of resources

Resources are the equipment, property or people needed to complete a task. Resources must be defined in the project management software. The availability of resources must also be identified. This allows the project manager to know when each resource can be used. Each resource is given a cost. A cost can be assigned to any resource. Giving a resource a cost allows the project manager to see how resource use affects the budget. Resources can be assigned to tasks. Each resource is given a number of hours or days it will be used for. This helps the project manager to know what resources are used when. This is very important when people are resources. If you allocate too many tasks to one person at the same time, they will not be able to complete these tasks. This leads to a person becoming stressed. They may rush tasks. This means that the task may not be completed to a good standard.

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Costings

Resources often have costs assigned to them. This allows the project manager to calculate the total cost of each task. The cost of a task is based on how many hours each resource is used for and how much each resource costs to use per hour. The project management software also records any extra expenses incurred by team members. The project management software will add these to the overall cost. The project management software can track how much is being spent over different time periods. For example, it could show graphs of the daily, weekly and monthly spending (see Figure 15.6).



Figure 15.6: Project management software tracks spending and displays it over set periods.

Project management software can report the total costs for individual resources or a set of resources. It can also export data about the project. It can export to common file types such as .xlsx or .csv. This means that other software (such as spreadsheet software) can then analyse the data.

Communication and collaborative working

Project management software has communication and collaborative working tools which help team members to **collaborate** during the project:

- **Calendars:** each team member has a calendar. The calendar shows what tasks the team member should be currently working on. Calendars can be synchronised with other third-party calendars, such as Google, iCal or Outlook. This allows team members to use mobile devices to check calendars rather than having to be at a desktop. This helps to ensure that each member's availability is up to date. Shared calendars enable meetings to be scheduled with each team member. Documents can be attached to these meetings so that all members of the meeting can review the documents before the meeting.
- **Instant messaging/video chat/video conferencing:** these tools enable team members to communicate in real time. They can use these tools to share ideas and discuss progress. Real-time communication is also useful if team members work in different countries.
- **Shared documents:** projects often use cloud storage for documents. This acts as a central store and allows team members to access the documents from wherever they may be. Documents can be assigned to tasks, resources or milestones. Changes to the documents can be tracked. Tracking changes allows each team member to see editing and changes to the document. Many documents also allow different versions to be saved or restored. Some storage management software emails team members to inform them when changes to documents have been made. Some project management software sends direct messages to a person's email when documents are changed or updated.
- **Discussions/forums:** these can be set up for tasks, documents or milestones. It allows team members to discuss tasks, documents or ideas. Forums are very useful when team members are not in the same time zone. Team members can see comments made

by other team members and reply. Forums can email notifications to team members when comments are made. They can also send alerts to project dashboards.

- **Progress:** the software informs team members and project managers of progress made on each task. Team members update tasks to show how close they are to finishing the task. The progress is fed into the project plan. When changes are made to the project timeline, automated email notifications are sent to all affected team members.

Decisions

Communication and decisions in project management software are usually recorded. This means that decisions can be tracked. Each decision can be linked to the person who made the decision and why that decision was made.

Project management software can also highlight potential issues. The project management software can suggest what changes need to be made. For example, it may suggest changing the timeline or allocating additional resources. These issues can be monitored to check if the issue has been resolved.

Project management software can compare differences between the project plan and current progress. The project management software can show how much time was spent on a task compared to how much time was planned for that task. This means that lessons can be learned for future tasks that are similar in nature.

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The project management software identifies a **critical path**. A critical path shows all tasks that must be completed if the project is to meet its deadline. Critical paths are monitored closely. It is important to identify potential delays in advance. Resources are diverted to critical tasks if they fall behind. This can help the project to meet its deadlines and recover any lost time.

Advantages and disadvantages of project management software

Table 15.1 sets out the advantages and disadvantages of project management software.

Advantages	Disadvantages
Can manage very large projects efficiently.	May be expensive to buy. If there are many users, several licences will be required – this will add to the cost.
Can produce a range of outputs for users to see how the project develops.	The business may not be able to afford to buy the software most appropriate for their requirements.
Can automatically update areas of the project if changes are made to another part.	Special training is often needed to use the software effectively. Additional training may be needed for different users.
Can send out automated reminders and updates to people.	Can be difficult to set up. This could lead to inaccurate results later.
Can be used to make calculations very quickly.	Team members may rely too much on software, and not realise that something is

Advantages	Disadvantages
	wrong if the software does not raise an alert.
Can analyse and make suggestions to projects to help save money and time.	Backups of the data need to be made in case of computer failure.
Can reduce the number of people required to manage the project. This can make communication between people and control of the project more effective.	Access and security levels will need to be set up properly to enable users to get access to the information they need.
Stakeholders can use the software for a summary of what stage the project has reach and any potential issues.	

Table 15.1: Advantages and disadvantages of project management software.

Questions

- 8 Explain how project management software can be used for allocating resources to tasks.
- 9 State the purpose of project management software.
- 10 Give two things that project management software can control or monitor.
- 11 Give two reasons why a company may choose not to use project management software.

ACTIVITY 15.01

Use the internet to research the following:

- What percentage of projects are successful?
- What percentage of projects run over budget?
- What percentage of projects in a company are likely to fail?
- What percentage of projects meet all their goals?

Write down ideas on how you would try to ensure that a project meets its goals, is delivered on time and stays in budget. What is the most important thing to consider for a project being successful? Discuss your findings with a classmate.